Surface-erosion metrics

Definitions exactly as computed by `ovito_insertion_movies.py`. Cycle i is the i-th He insertion; $i = 0$ is the equilibrated, pre-impact slab.

The three reported quantities

Growth — amorphous-layer thickness

$$G(i) \;=\; z_{\mathrm{surf}}(i) \;-\; z_{\mathrm{cryst}}(i) \;\geq\; 0 \tag{1}$$

Distance from the top of the material down to the top of the ordered lattice.

Regression (order deficit)

$$R_D(i) \;=\; -\big[\, D(i) - D(0) \,\big], \qquad D \;=\; \int_{z_{\mathcal{B}}}^{\infty} \varphi(z)\,\theta(z)\,\mathrm{d}z \tag{2}$$

Thickness of equivalent *fully disordered* material gained since cycle 0, signed so that lost order is negative. No threshold, so no resolution floor.

Regression (ordered-lattice front)

$$R(i) \;=\; z_{\mathrm{cryst}}(i) \;-\; z_{\mathrm{cryst}}(0) \;\leq\; 0 \tag{3}$$

How far the top of the ordered lattice has moved down since cycle 0.

For a sharp interface between a fully amorphous layer and perfect crystal, $R_D \approx R$. They part company when the disorder is partial or graded: R is a threshold crossing and stays pinned at the surface until the order parameter actually crosses $f_S S_{\mathrm{b}}$, whereas R_D responds in proportion to how disordered the near-surface material is.

Ingredients

Atoms. All W and C atoms (He excluded) whose lateral position lies inside the impacted column, of area

$$A_{\mathrm{imp}} = (fL_x)(fL_y), \qquad f = \tfrac{2}{3}, \tag{4}$$

with heights z_j. The un-hit outer frame is excluded so it does not dilute either metric.

Periods. From the cycle-0 bulk band $\mathcal{B}$ (below), the layer period d puts every atomic plane in phase, and the chemical period c does the same for the W sublattice alone:

$$d = \arg\max_{d'} \left| \left\langle e^{2\pi i z_j / d'} \right\rangle_{j \in \mathcal{B}} \right|, \qquad c = \arg\max_{c'} \left| \left\langle e^{2\pi i z_j / c'} \right\rangle_{j \in \mathcal{B},\,\mathrm{W}} \right|. \tag{5}$$

For basal (0001), $d = c_{\mathrm{WC}}/2 \approx 1.42\,\text{Å}$ and $c = c_{\mathrm{WC}} \approx 2.83\,\text{Å}$.

Density. Atom count in a sliding window one chemical period wide, which averages the alternating W/C planes out exactly:

$$\bar{\rho}(z) = \frac{1}{c} \sum_j \mathbf{1}\big[\, |z_j - z| \le \tfrac{c}{2} \,\big] . \tag{6}$$

Layering order parameter.

$$\tilde{S}(z) = \left| \frac{1}{N_w(z)} \sum_{|z_j - z| \le w} e^{2\pi i z_j / d} \right|, \qquad w = 2.5\,\text{Å}, \tag{7}$$

where $N_w(z)$ is the number of atoms in the window. $\tilde{S}$ is scored only where $N_w(z) \ge \max\big(N_{\min},\ f_{\text{occ}} \max_z N_w\big)$ ($N_{\min} = 40$, $f_{\text{occ}} = 0.12$), and $S(z)$ is $\tilde{S}$ after a 3-point boxcar on the grid $z_k = z_{\mathcal{B}} + k\,\Delta z$, $\Delta z = 0.25\,\text{Å}$.

Bulk references. Over the thermalised band $\mathcal{B} = [\, z_{\text{bot}} + 20\,\text{Å},\ z_{\text{bot}} + 45\,\text{Å}\,]$ of the same frame, with $z_{\mathcal{B}}$ its top:

$$\rho_{\text{b}} = \operatorname*{median}_{z \in \mathcal{B}} \bar{\rho}(z), \qquad S_{\text{b}} = \operatorname*{median}_{z \in \mathcal{B}} S(z). \tag{8}$$

Normalising by the same frame's bulk absorbs Debye–Waller damping.

Surface. The Gibbs dividing surface, linearly interpolated between grid points:

$$z_{\text{surf}} = \max\big\{\, z : \bar{\rho}(z) \ge f_\rho\, \rho_{\text{b}} \,\big\}, \qquad f_\rho = \tfrac{1}{2}. \tag{9}$$

Top of the ordered lattice. Walking up from the bulk, the ordered region ends at the first run of n consecutive sub-threshold bins:

$$k^* = \min\big\{\, k \ge 1 \ :\ S(z_{k+m}) < f_S\, S_{\text{b}} \ \ \forall\, m = 0, \ldots, n-1 \,\big\}, \qquad f_S = \tfrac{1}{2},\ n = 3. \tag{10}$$

The crossing is interpolated inside the last ordered bin and capped at the surface,

$$\hat{z} = z_{k^*-1} + \frac{S(z_{k^*-1}) - f_S S_{\text{b}}}{S(z_{k^*-1}) - S(z_{k^*})}\, \Delta z, \qquad z_{\text{cryst}} = \min\big(\hat{z},\ z_{\text{surf}}\big), \tag{11}$$

and $z_{\text{cryst}} = z_{\text{surf}}$ when no such run exists (ordered right up to the surface). On a pristine slab this gives $G = 0$ exactly.

Order deficit integrand.

$$\varphi(z) = \begin{cases} \min\big(1 - S(z)/S_{\text{b}},\ 1\big) & \text{where } S \text{ is scored,} \\ 1 & \text{otherwise (e.g. loose adatoms),} \end{cases} \qquad \theta(z) = \min\Big(\max\big(\bar{\rho}(z)/\rho_{\text{b}},\ 0\big),\ 1\Big). \tag{12}$$

θ makes vacuum contribute nothing. φ is deliberately *not* clipped at zero, so thermal scatter of S about S_{b} in ordered material averages out instead of accumulating. In practice $D = \Delta z \sum_{k \ge 0} \varphi(z_k)\, \theta(z_k)$.

Plotting, time and fluence

Despiking. Curves drawn in the erosion figures are a centred median over $w_{\mathrm{med}} = 5$ cycles, which removes one-frame threshold excursions without moving a genuine step:

$$\tilde{y}_i = \mathrm{median}\{\, y_{i-2}, \ldots, y_{i+2} \,\}. \tag{13}$$

`erosion.csv` always holds the unfiltered values.

Time and fluence. One ion per cycle, at insertion interval T_c, over the impacted area:

$$t_i = i\,T_c \quad (T_c = 20\,\mathrm{ps}), \qquad \Phi_i = \frac{i}{A_{\mathrm{imp}}} = \frac{i}{A_{\mathrm{imp}}[\mathrm{\AA}^2] \times 10^{-16}}\ \mathrm{ions\,cm^{-2}}. \tag{14}$$

Symbols

Symbol	Meaning	Config name
d	layer period (every plane in phase)	`layer_period_A`
c	chemical period (W sublattice)	`chem_period_A`
w	order-parameter window half-width	`WIN_HALF`
Δz	profile grid spacing	`BIN_DZ`
$\mathcal{B}$	bulk reference band	`BULK_BAND_A`
f_ρ	density fraction defining z_{surf}	`RHO_FRACTION`
f_S	order fraction still counted as crystal	`S_THRESHOLD`
n	sub-threshold bins that end the lattice	`ORDER_RUN_BINS`
N_{min}, f_{occ}	window statistics floor	`MIN_WIN_ATOMS`, `MIN_WIN_OCC`
w_{med}	median despike width (plots only)	`DESPIKE_WIN`
T_c	insertion-to-insertion interval	`T_CYCLE_PS`
f	impacted fraction of each lateral side	`IMPACT_FRAC`